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Experiment No. 05 Dynamic Routing 3–
Path Vector Routing Protocols Border
Gateway Protocol (BGP)

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Abstract

This experiment aims to provide hands-on introduction to network design and configuration using Cisco Packet Tracer. The main goals were to develop and test a simulated network topology, and understand how IGP (OSPF) and EGP (BGP) work each by itself, and how they work with each other where essential networking concepts-such as IP addressing, and routing protocol-were applied in practice. Output included configuring routers, switches, establishing both OSPF and BGP sessions, and ensuring connectivity between multiple PCs using commands like *ping*, *show ip route*, and *show ip bgp summary*. In addition, packet sniffing tools -like the sniffer- were used to capture and analyze BGP message types (OPEN, UPDATE, KEEPALIVE, and NOTIFICATION). Finally, it reinforced both my theoretical concepts and practical troubleshooting skills.

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Acronyms and Abbreviations

<i>OSPF</i>	<i>Open Shortest Path First</i>
<i>BGP</i>	<i>Border Gateway Protocol</i>
<i>LSDB</i>	<i>Link State Database</i>
<i>AS</i>	<i>Autonomous System</i>
<i>IGP</i>	<i>Interior Gateway Protocol</i>
<i>EGP</i>	<i>Exterior Gateway Protocol</i>

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1 Introduction

“Dynamic routing enables routers to automatically exchange routing information and update their routing tables without manual configuration. Dynamic routing protocols are classified into two main types: Interior Gateway Protocols (IGPs) and Exterior Gateway Protocols (EGPs).” [1]

IGPs -like Open Shortest Path First (OSPF) are used inside one Organization\Autonomous Systems (AS) whereas “EGPs facilitate communication between different ASes. The most widely used EGP is the Border Gateway Protocol (BGP). Unlike IGPs, which prioritize internal efficiency, BGP is designed for scalability, stability, and policy-based routing, making it essential for Internet-wide routing. Note that the Internet is composed of multiple interconnected ASes, each managed by different organizations. Figure 1-1 illustrates the classification of IGPs and EGPs.” [1]

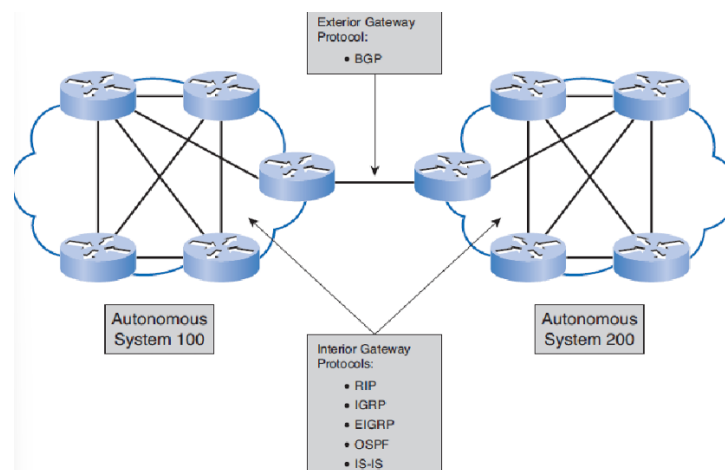


Figure 1-1: AS connectivity using BGP links.[1]

1.1 Border Gateway Protocol (BGP): Intro.

1.1.1 definition

BGP is a Path Vector Protocol designed for inter-domain routing. It makes routing decisions based on a set of attributes. The most critical of these is the AS Path, which records the sequence of ASes a route has traversed. Each AS is assigned a globally unique 16-bit AS number by the Internet Assigned Numbers Authority (IANA). [1]

1.1.2 BGP's different relations

BGP sets up peer relationships between routers (BGP speakers) to share routing information. Peers are classified as:

- iBGP (Internal BGP): Connects routers within the same AS (Administrative Distance: 200). [1]
- eBGP (External BGP): Connects routers within different ASes (Administrative Distance: 20). [1]

1.2 BGP features:

- 1- **Inter-Domain Routing:** BGP is designed to enable inter-AS routing, so it is a crucial component of Internet connectivity. Its design is powerful enough to support interconnections across various networks, ranging from small businesses to large international carriers. [1]
- 2- **Policy-Based Routing:** BGP allows precise control over the selection of routes. Network administrators can define routing policies to select which routes are preferable in order to balance the load and offer redundancy. [1]
- 3- **Fault Tolerance and Redundancy:** With BGP, organizations can establish multiple peering relationships with various external networks. This redundancy is necessary to maintain continuous connectivity even if a single link is down. [1]
- 4- **“Route Selection:** BGP selects the optimal path based on the AS Path and other attributes. Since AS Path information is exchanged between peers, routers can make informed decisions on the best route to a destination AS.” [1]
- 5- **“Prevention of Routing Loops:** When a router receives a routing update, it checks the AS Path for its own AS number. If found, the route is discarded, preventing loops where data would circulate endlessly between ASes.” [1]

1.3 BGP Attributes:

1. AS Path: Shows ASes traversed by the route (loop avoidance).[2]
2. Next Hop: Indicates the next-hop IP address of the route.[2]
3. Local Preference: Influences outbound traffic (higher values are preferable; iBGP-only).[2]
4. MED (Multi-Exit Discriminator) recommends optimal entry points to adjacent Autonomous Systems.[2]
5. Weight (Cisco-specific): Locally significant metric for preferring routes.[2]

1.4 BGP Session Establishment:

1. Idle: No connection active; starting state. [1]
2. Connect: Forms a TCP connection (port 179) with the peer. [1]
3. Active: Waits for incoming TCP connections if the initial attempt fails. [1]
4. OpenSent: Sends an OPEN message and awaits the response from the peer. [1]
5. OpenConfirm: Validates OPEN message parameters and awaits KEEPALIVE. [1]
6. Established: Peers share UPDATE messages to distribute routing information. [1]

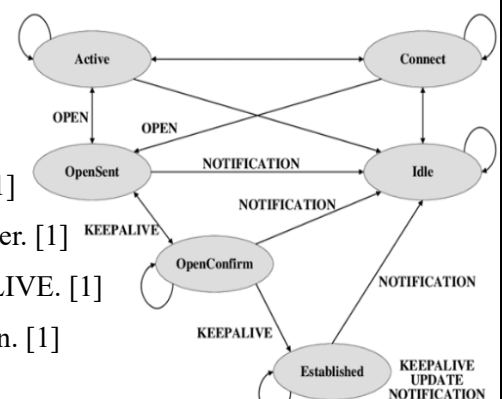


Figure 1-2: BGP Session State

2 Procedure and Discussion

2.1 Setting up the Topology

2.1.1 Topology Diagram & Components

The network was designed according to the topology illustrated in **Figure 2-1**, where the variable x represents the last two digits of my student ID (i.e., 82).

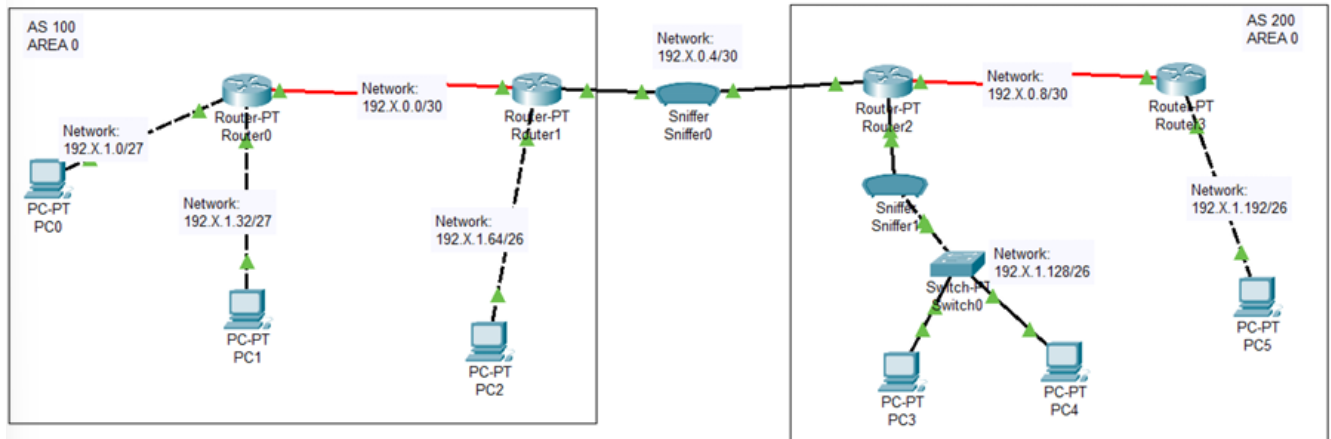


Figure 2-1: Topology. [1]

Table 2-1 shows the devices and connections implemented in the simulation:

Table 2-1: Used Devices

Routers	Switches	PCs	Sniffers	Connections
Router-PT	Switch-PT	PC-PT	Sniffer	Automatic connections (start from the routers to avoid problems)

2.1.2 IPs Distribution

The network topology in **Figure 2-1** defines the first three octets of the IP addresses for all devices, while the fourth octet (host portion) is assigned according to the numbering convention specified in **Figure 2-2**.

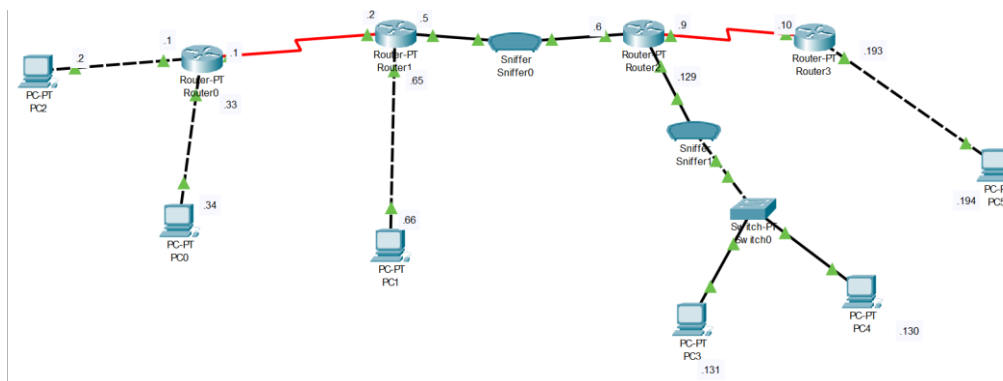


Figure 2-2: IP Distribution.

2.1.3 IP Configuration

2.1.3.1 Routers Configuration

In this section we will see how we can assign IP addresses for routers.

In general, there are two ways to configure them; either by GUI or the CLI. I will show GUI configuration for the first 2 routers then we will see how to do the same in CLI for every router. Note that in CLI for every router, we should first go to the privileged mode by inserting `enable` then enter. Then we should go to the global configuration mode by `config t`. After that, for every interface you should type:

```
Interface <TYPE> <SLOT> / <PORT>
ip address <IP ADDRESS> <MASK>
no shutdown
```

[1]

Figure 2-3 illustrates how to configure Router0 using GUI

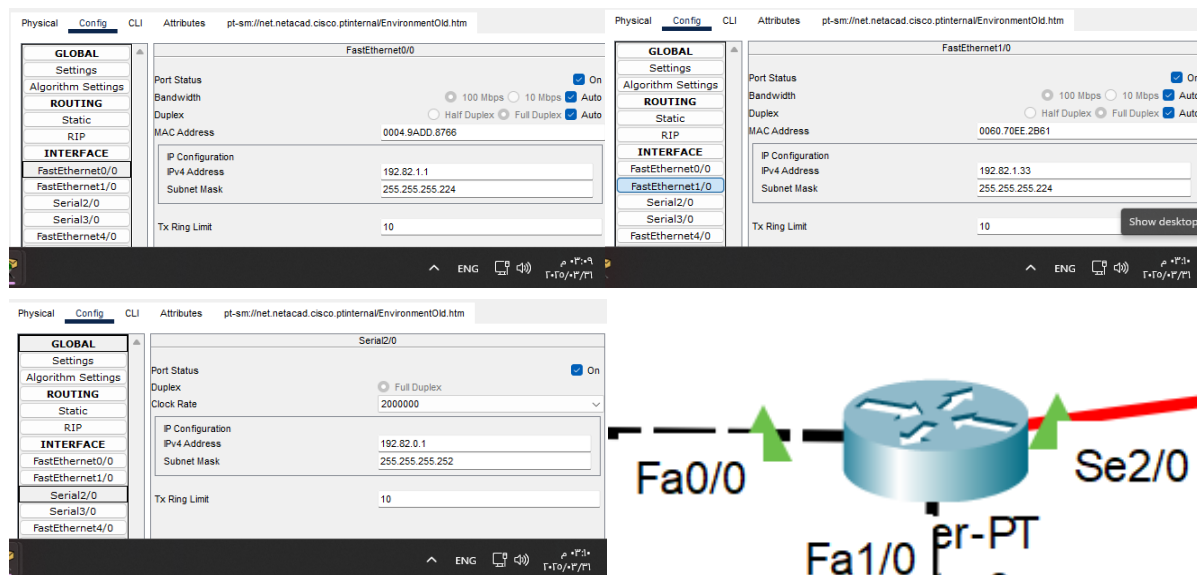


Figure 2-3: R0 Configuration

Or we can configure Router0 as follows:

```
interface FastEthernet0/0
ip address 192.82.1.1 255.255.255.224
no shutdown
exit
interface FastEthernet1/0
ip address 192.82.1.33 255.255.255.224
no shutdown
exit
interface Serial2/0
```

```
ip address 192.82.0.1 255.255.255.252
no shutdown
exit
```

for Router1:

Figure 2-4 illustrates how to configure it using GUI

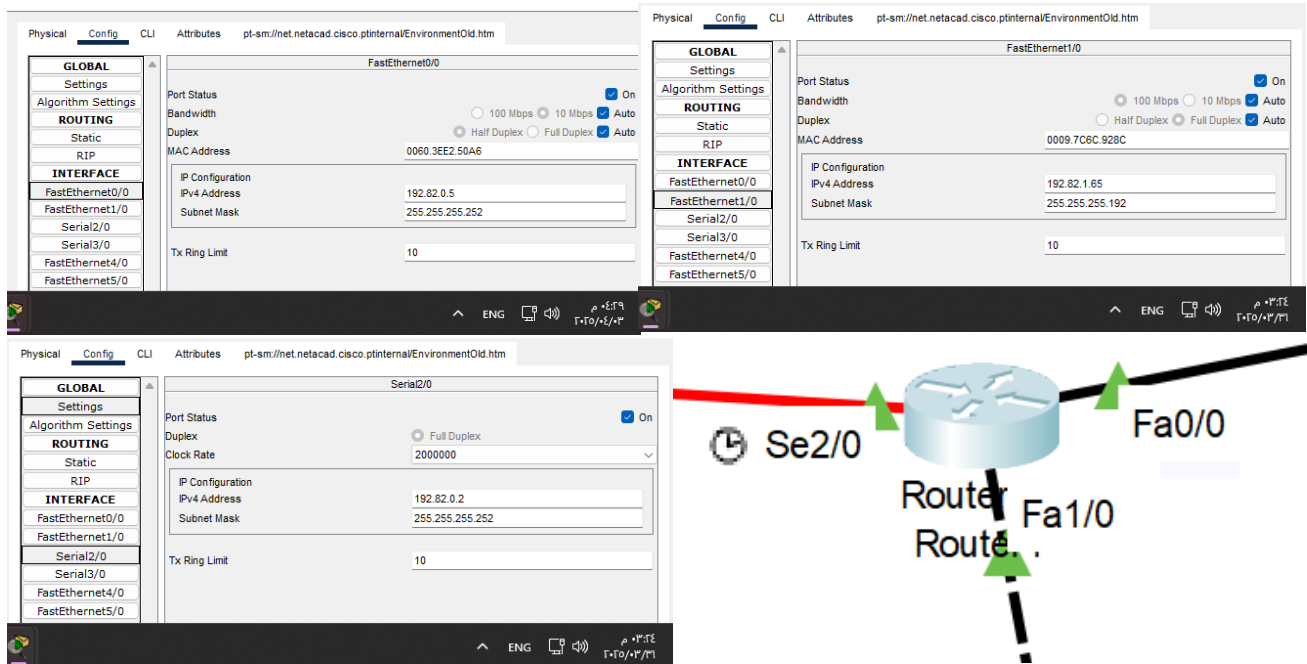


Figure 2-4: R1 Configuration

By CLI:

```
interface FastEthernet0/0
ip address 192.82.0.5 255.255.255.252
no shutdown
exit
interface FastEthernet1/0
ip address 192.82.1.65 255.255.255.192
no shutdown
exit
interface Serial2/0
ip address 192.82.0.2 255.255.255.252
no shutdown
exit
```

For Router2:

```

interface FastEthernet0/0
ip address 192.82.0.6 255.255.255.252
no shutdown
exit
interface FastEthernet1/0
ip address 192.82.1.129 255.255.255.192
no shutdown
exit
interface Serial2/0
ip address 192.82.0.9 255.255.255.252
no shutdown
exit

```

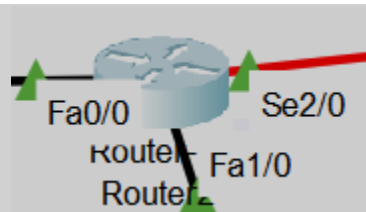


Figure 2-5: Router2 Interfaces

For Router3:

```

interface FastEthernet0/0
ip address 192.82.1.193 255.255.255.192
no shutdown
exit
interface Serial2/0
ip address 192.82.0.10 255.255.255.252
no shutdown
exit

```

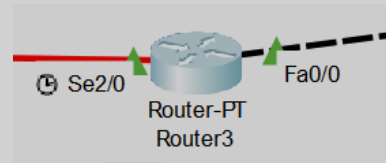


Figure 2-6: Router3 Interfaces

2.1.3.2 PCs Configuration

Figures 2-7,2-8,2-9 show how to configure IP addresses on every PC

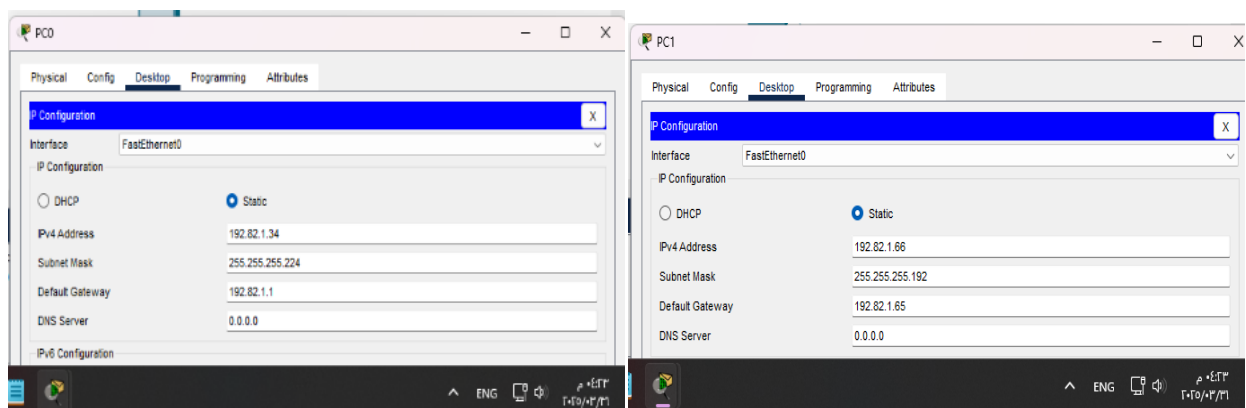


Figure 2-7: IP Configuration for PC0 & PC1

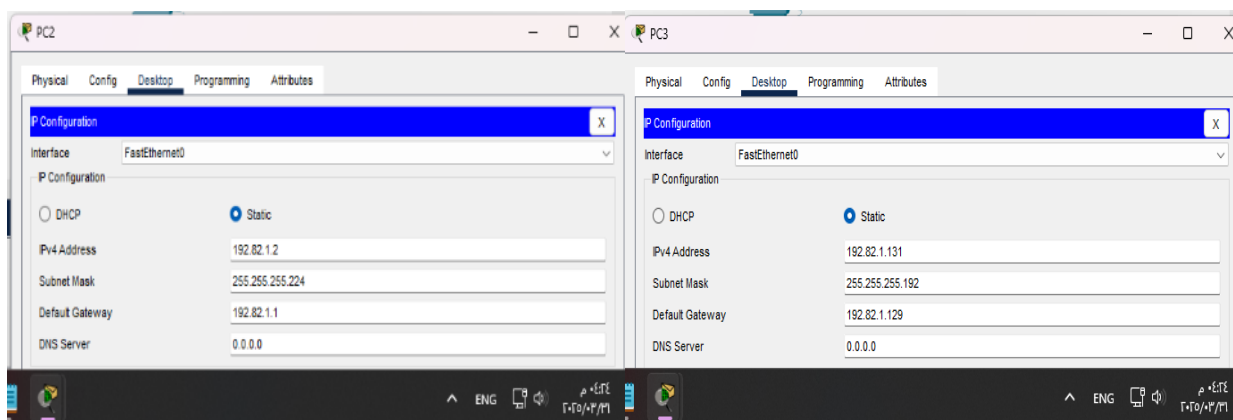


Figure 2-8: IP Configuration for PC2 & PC3

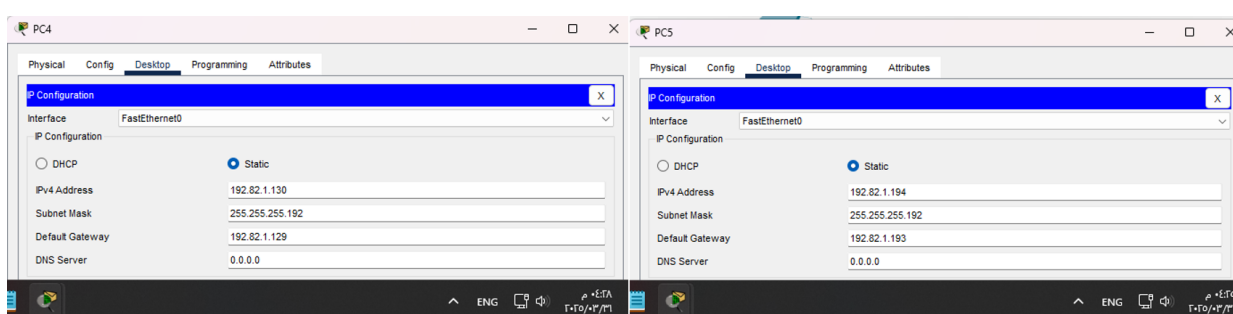


Figure 2-9: IP Configuration for PC4 & PC5

2.2 IGP (OSPF) Configuration

After IP configuration, we should configure OSPF for every router as follows:

- We will run the OSPF process on every router by commanding [1] `router ospf <PROCESS-ID>`
- We can use any number as a process-id, but we will use 1
- Now, every network connected to this router except anyone connected to different autonomous systems should be added to the OSPF process.
- We can use the command [1] `network <NETWORK-ID> <WILDCARD-MASK> area <AREA-ID>`

Table 2-2 outlines the methodology for configuring OSPF routing across the network topology.

Table 2-2: networks added for each router

Router Name	Networks Added (Commands in the OSPF process)
Router0	network 192.82.1.0 0.0.0.31 area 0 network 192.82.1.32 0.0.0.31 area 0 network 192.82.0.0 0.0.0.3 area 0
Router1	network 192.82.0.0 0.0.0.3 area 0 network 192.82.1.64 0.0.0.63 area 0 network 192.82.0.4 0.0.0.3 area 0

Router2	network 192.82.0.8 0.0.0.3 area 0 network 192.82.1.128 0.0.0.3 area 0
Router3	network 192.82.0.8 0.0.0.3 area 0 network 192.82.1.192 0.0.0.63 area 0

Now, by this we have finished configuring OSPF, but we should wait a little before testing until building the LSDB.

2.3 EGP (BGP) Configuration

Configuring BGP is straightforward. The configuration steps are as follows:

- For every router that resides between two Autonomous Systems (ASes), we need to enable BGP by specifying the AS number using the following command: `router bgp <AS-NUMBER>` where AS-NUMBER is for the router itself
- Inside the process of bgp we must establish peer relationships with neighboring routers in different ASes (eBGP) To do this, we use the command: `neighbor <NEIGHBOR-IP> remote-as <AS-NUMBER>` where AS-NUMBER is for the neighbor router.
- We can configure BGP Timers(optional) such as keepalive and hold time timers by the command `timers bgp <keepalive> <hold time>`. It is preferred to make hold time = 3 times of keepalive time. I want to note that their default values are 180, 60 respectively.

Table 2-3 outlines the methodology for configuring BGP routing across the network topology.

Table 2-3: BGP Configuration

Router Name	Networks Added (Commands in the global config. Mode sequentially)
Router1	router bgp 100 neighbor 192.82.0.6 remote-as 200 exit timers bgp 30 90
Router2	router bgp 200 neighbor 192.82.0.5 remote-as 100

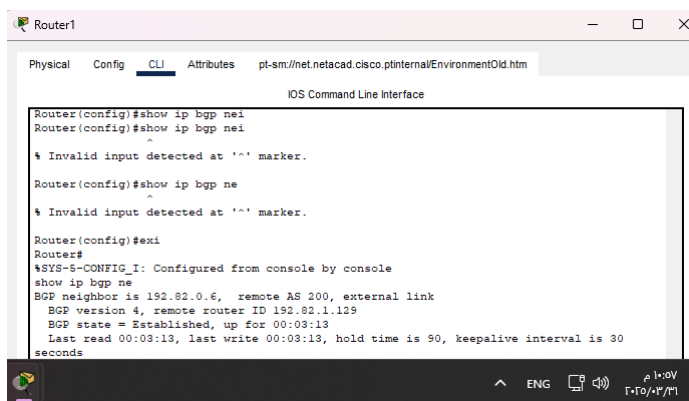


Figure 2-10: hold time and keepalive time after configuration

As we see, for router1 the hold time and keepalive time were changed to 90 30 which is what we did in the commands above.

2.4 Distributing routing protocols on each other

2.4.1 Distributing BGP over OSPF

“To allow the OSPF to communicate with the BGP, a redistribute command is used to define the BGP protocol over the OSPF protocol. That is:” [1]

```
router ospf <PROCESS —ID>
redistribute bgp <AS—NUMBER> subnets
```

[1]

We will apply this command for Router1 and Router2 as shown in Table 2-4:

Table 2-4: Enabling BGP over OSPF

Router Name	Commands in global configuration
Router1	<pre>router ospf 1 redistribute bgp 100 subnets bgp</pre>
Router2	<pre>router ospf 1 redistribute bgp 200 subnets</pre>

2.4.2 Distributing OSPF over BGP

“To allow the BGP to communicate with the OSPF a redistribute command is used to define the OSPF protocol over the BGP protocol:” [1]

```
router bgp <AS—NUMBER>
redistribute ospf <PROCESS —ID>
```

[1]

We will apply this command for Router1 and Router2 as shown in Table 2-5:

Table 2-5: Enabling OSPF over BGP

Router Name	Commands in global configuration
Router1	router bgp 100 redistribute ospf 1
Router2	router bgp 200 redistribute ospf 1

2.5 Sniffers Configuration

To enable packet capture, follow these steps:

- Navigate to Sniffer → GUI → Service. [1]
- Toggle Check On/Off to enable or disable packet capturing.[1]
- Select Port 0 or Port 1 to choose which interface captures incoming packets, allowing you to control the direction and source of traffic.[1]
- Instead of letting the sniffer capture all the packets, we can apply some filters to show specific types of messages. For this experiment, we will use OSPF, BGP, and ICMP – which is used for ping. Figure 2-11 shows sniffer configuration.

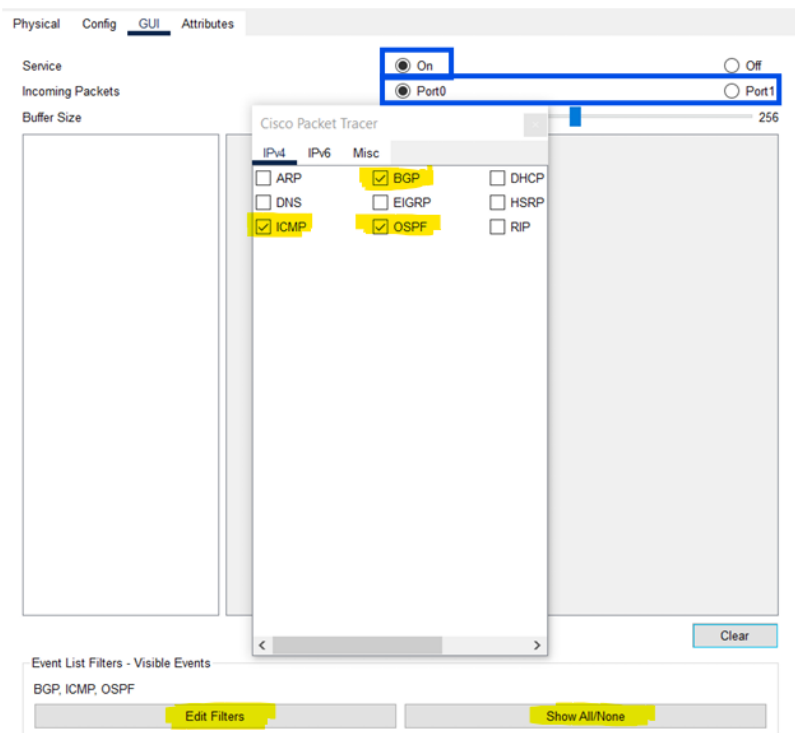


Figure 2-11: Sniffer configuration

3 Results

3.1 BGP Routing and Neighbor Information

We can inspect the BGP configuration and monitor its behavior using the command:

Show ip route

Figures 3-1 and 3-2 show the application for that command:

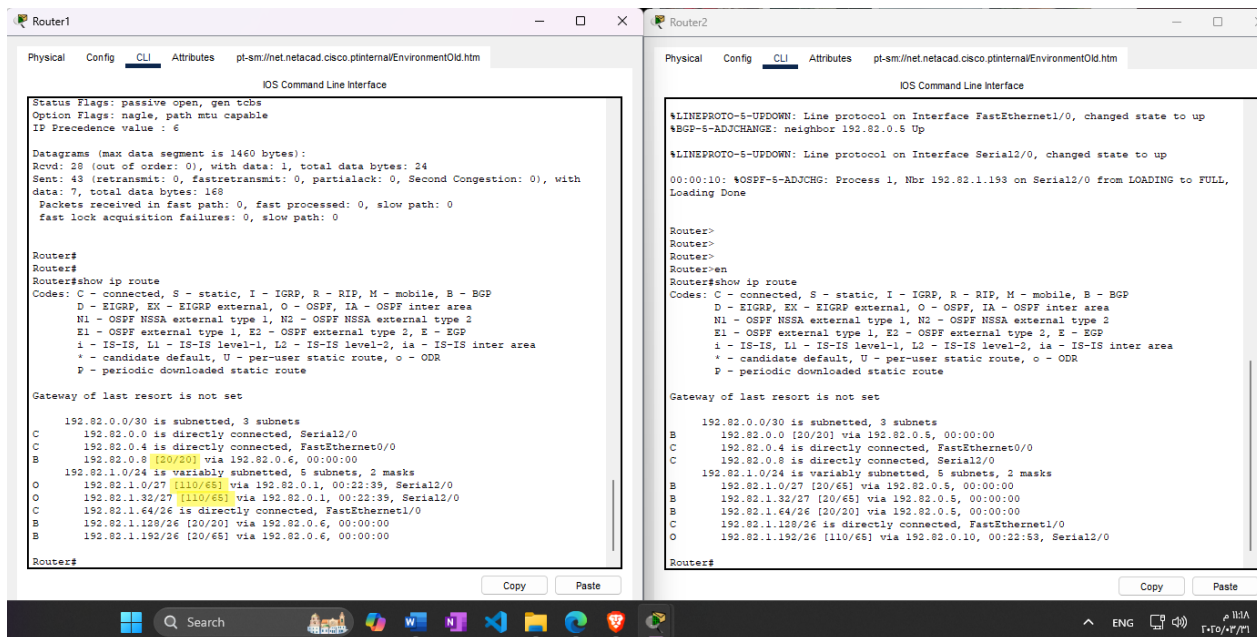


Figure 3-1: routes information for BGP routers

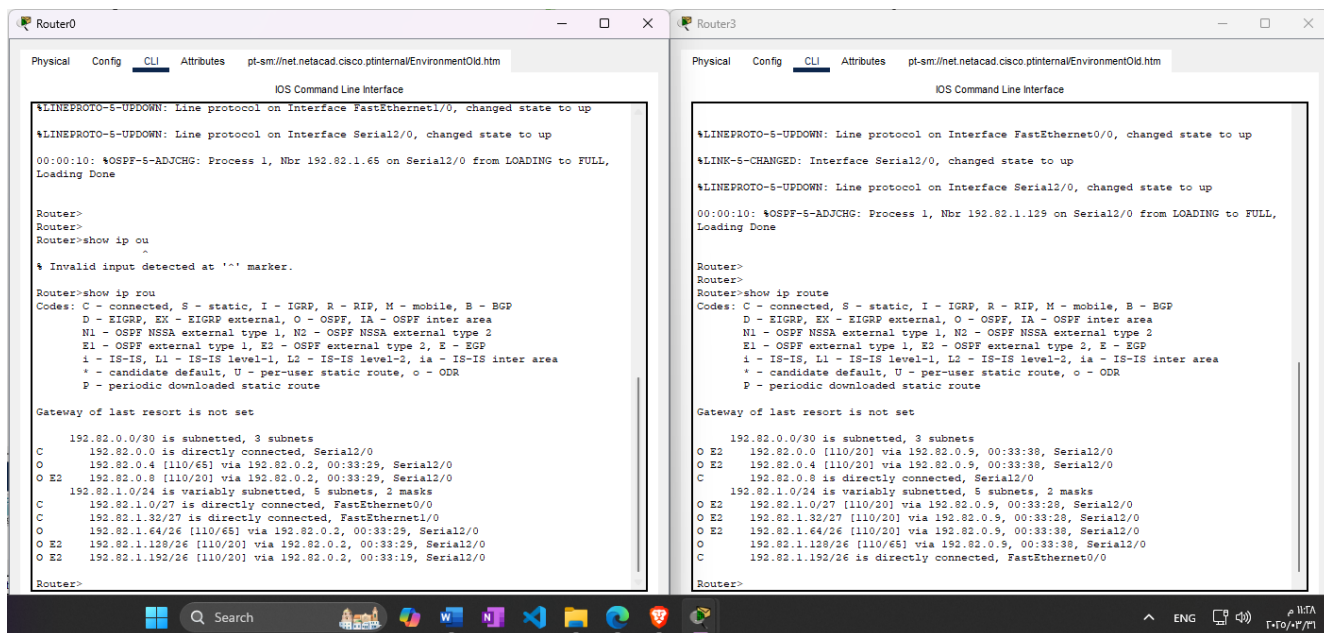


Figure 3-2: routes information for non-BGP routers

Note here that E2 flag means that the route comes from the distribution between both protocols. O means that the route is connected by OSPF, and B means that the route is connected by BGP. The brackets containing two numbers on the form [a/b] refer to the [administrative distance/cost(metric) of routing protocol] respectively. This command provides information about which routes are reachable, how they are reached, through which interface and next-hop router, how the route was discovered, and the associated cost. reaching them

Another applicable command is:

```
show ip bgp summary
```

[1]

Which shows a brief for all the details that are related to the BGP process.

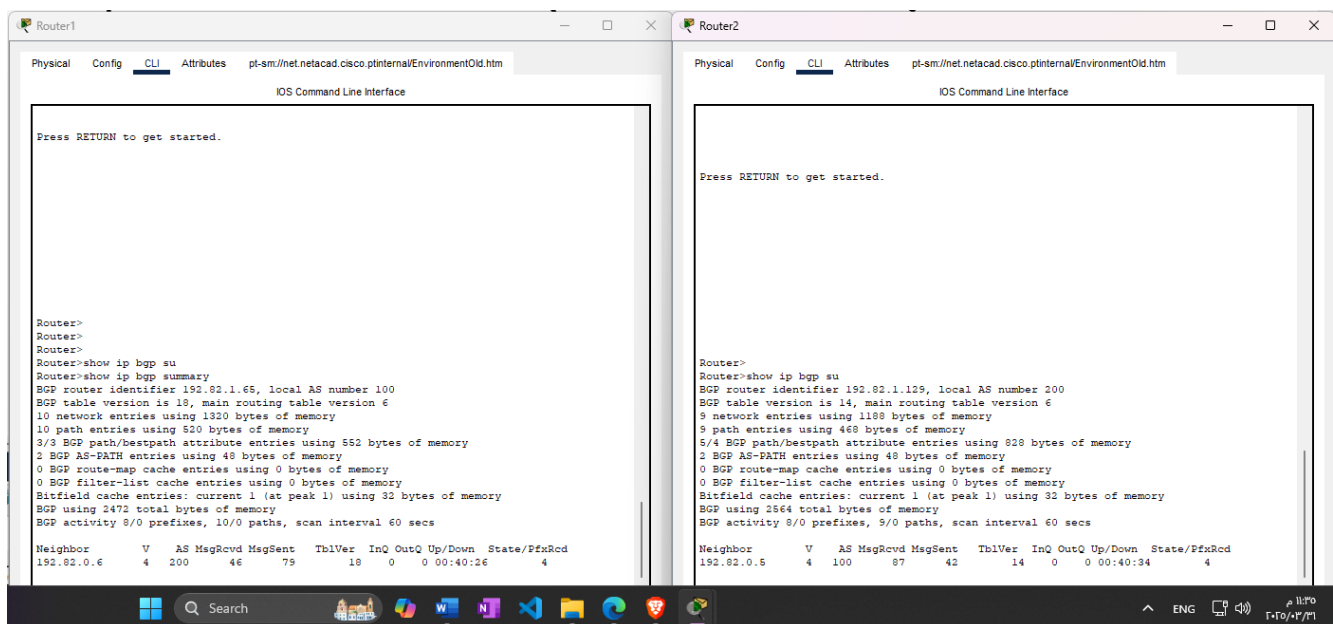


Figure 3-3: The *show ip bgp sum* command application

From Figure 3-3 we can monitor essential details such as the BGP router ID, local AS number, neighbor IP addresses, their AS numbers, the number of prefixes received, and the state of each BGP session.

3.2 Packet Sniffing

3.2.1 Captured BGP Messages

BGP have a lot of message types go to 6.2 Appendix B to see more. Figure 3-4 shows the BGP open message, whereas Figure 3-5 shows the BGP update message.

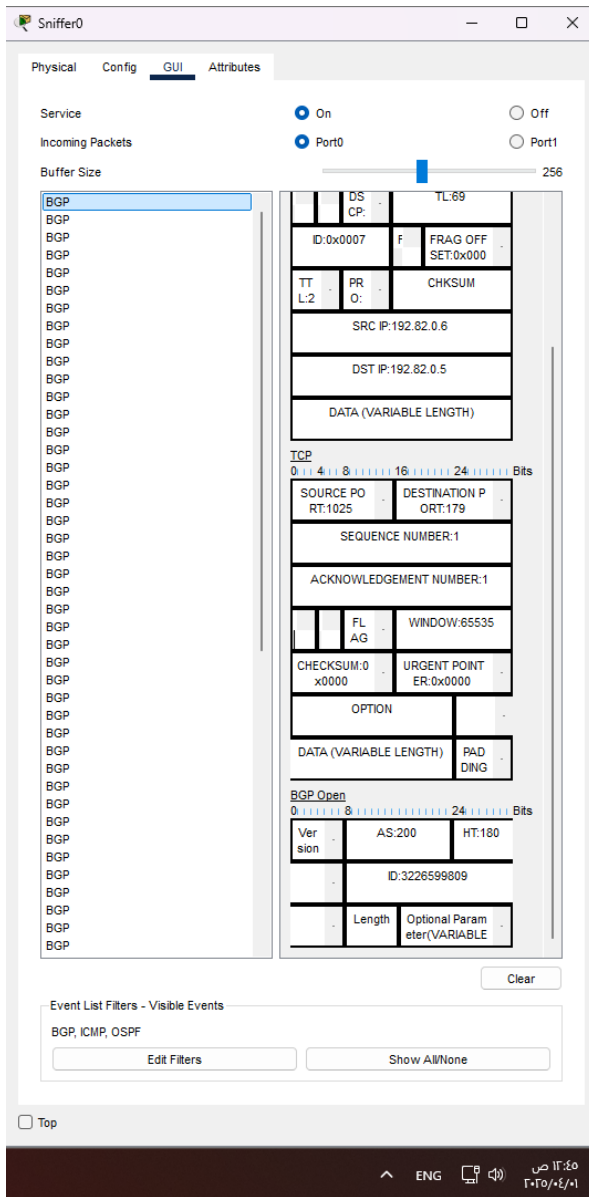


Figure 3-4: BGP open message

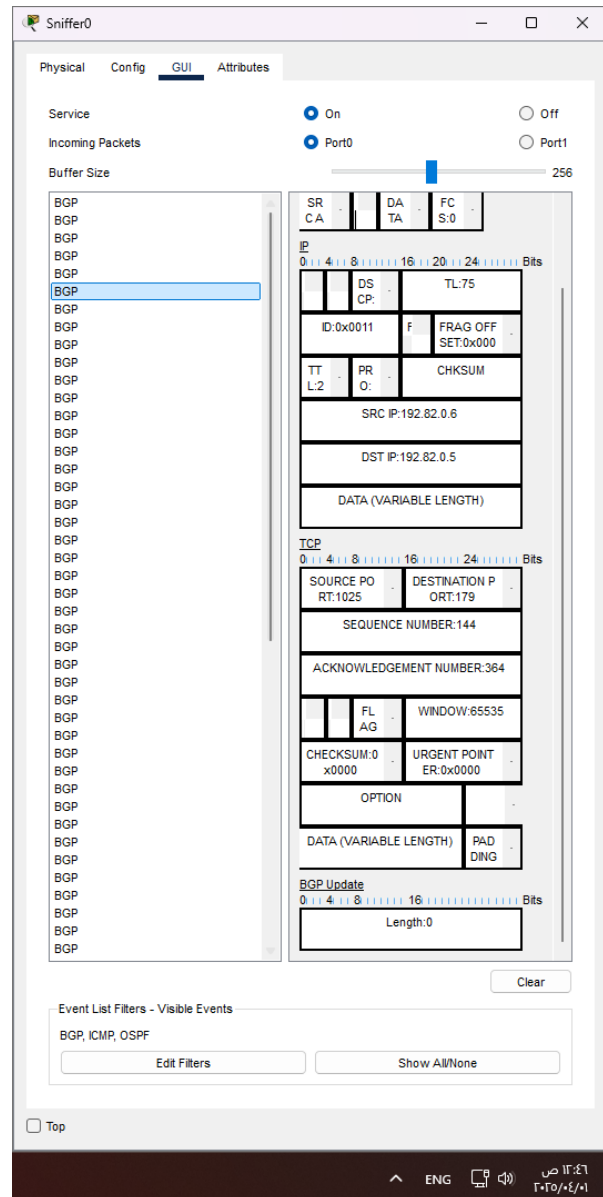


Figure 3-5: BGP Update message

Figure 3-5 shows the BGP keepalive message, whereas Figure 3-6 shows the BGP notification message.

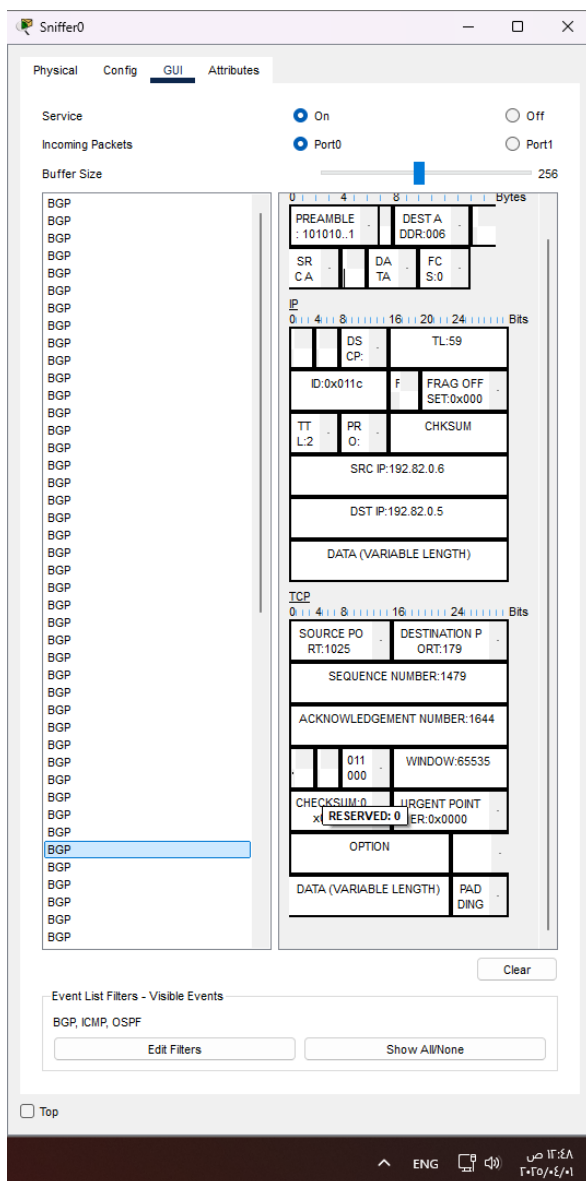


Figure 3-6: BGP keepalive message.

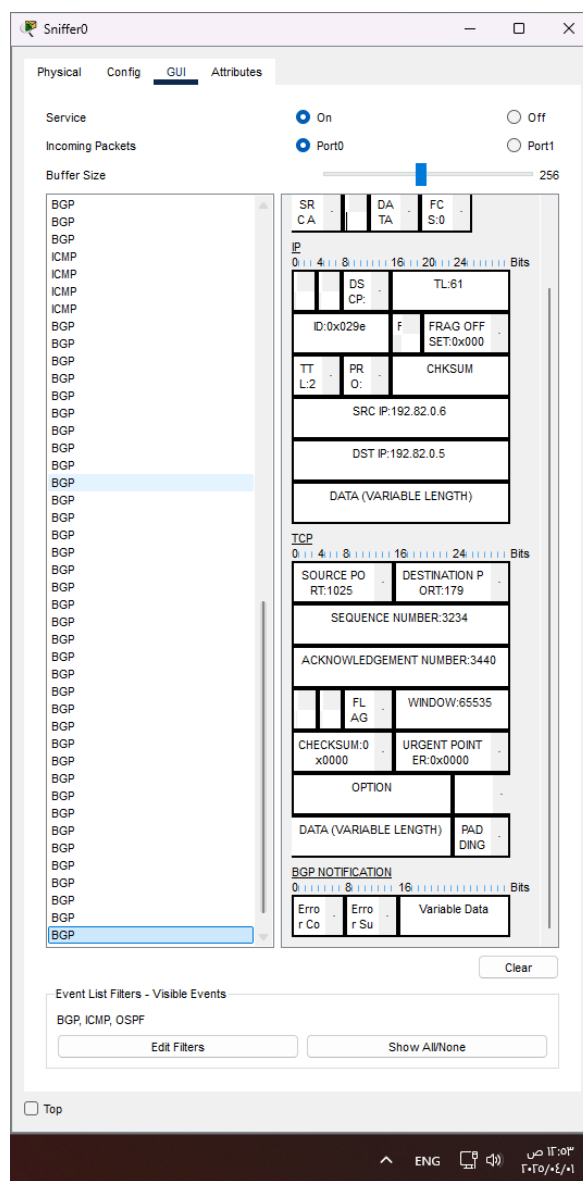


Figure 3-7: BGP notification message.

3.2.2 Captured OSPF & ICMP Messages

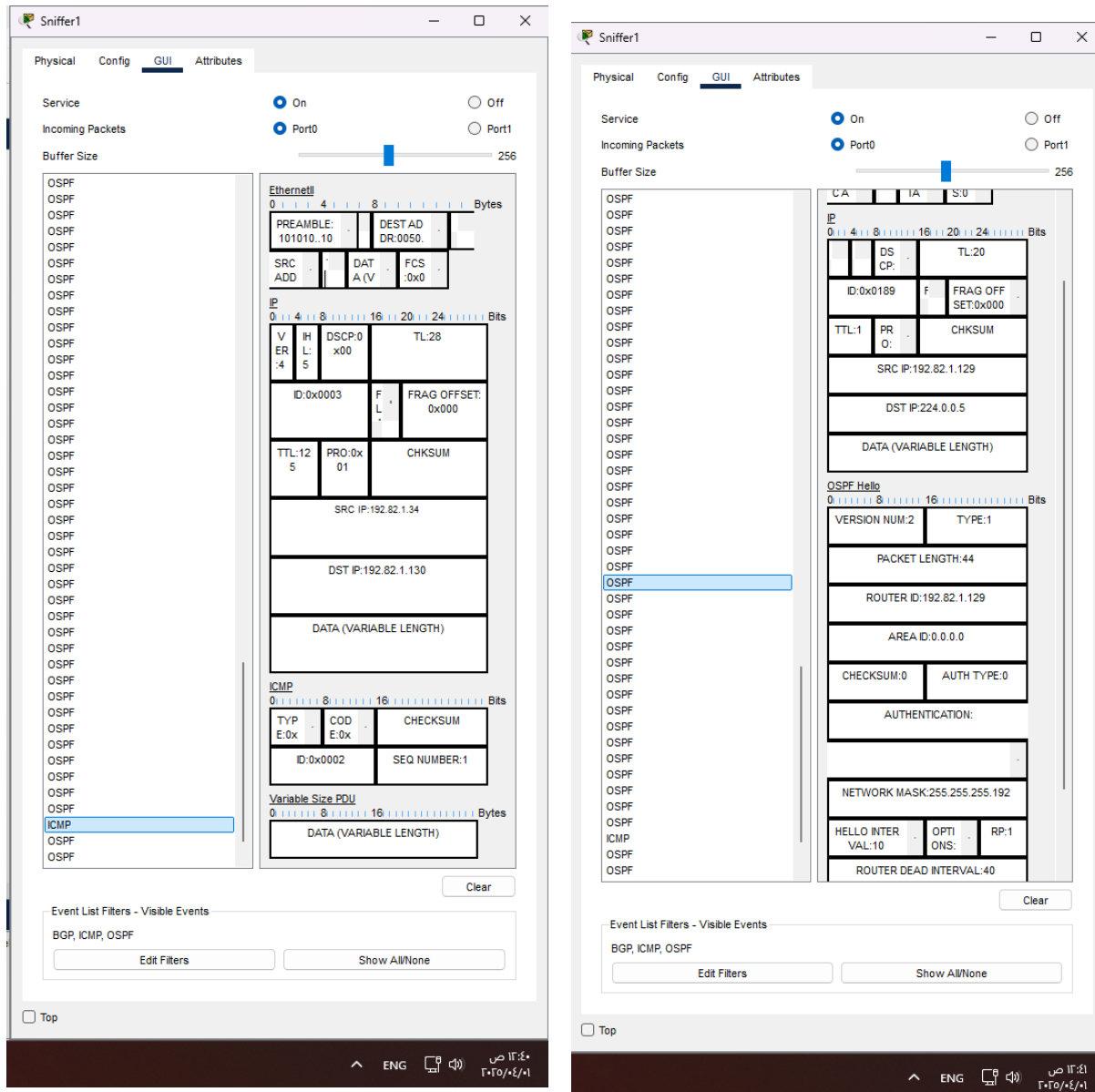


Figure 3-8: ICMP message from pc4 to pc2, and OSPF open message

Figure 3-8 shows ICMP and OSPF messages on sniffer 1 and the information sent about protocols.

3.3 Connectivity Between Devices

I tested connectivity between every host network from every pc using the ping command. Figure 3-9 illustrates the connection from PC0 to PC2 and from PC0/PC2 to PCs 1, 4, 5 respectively

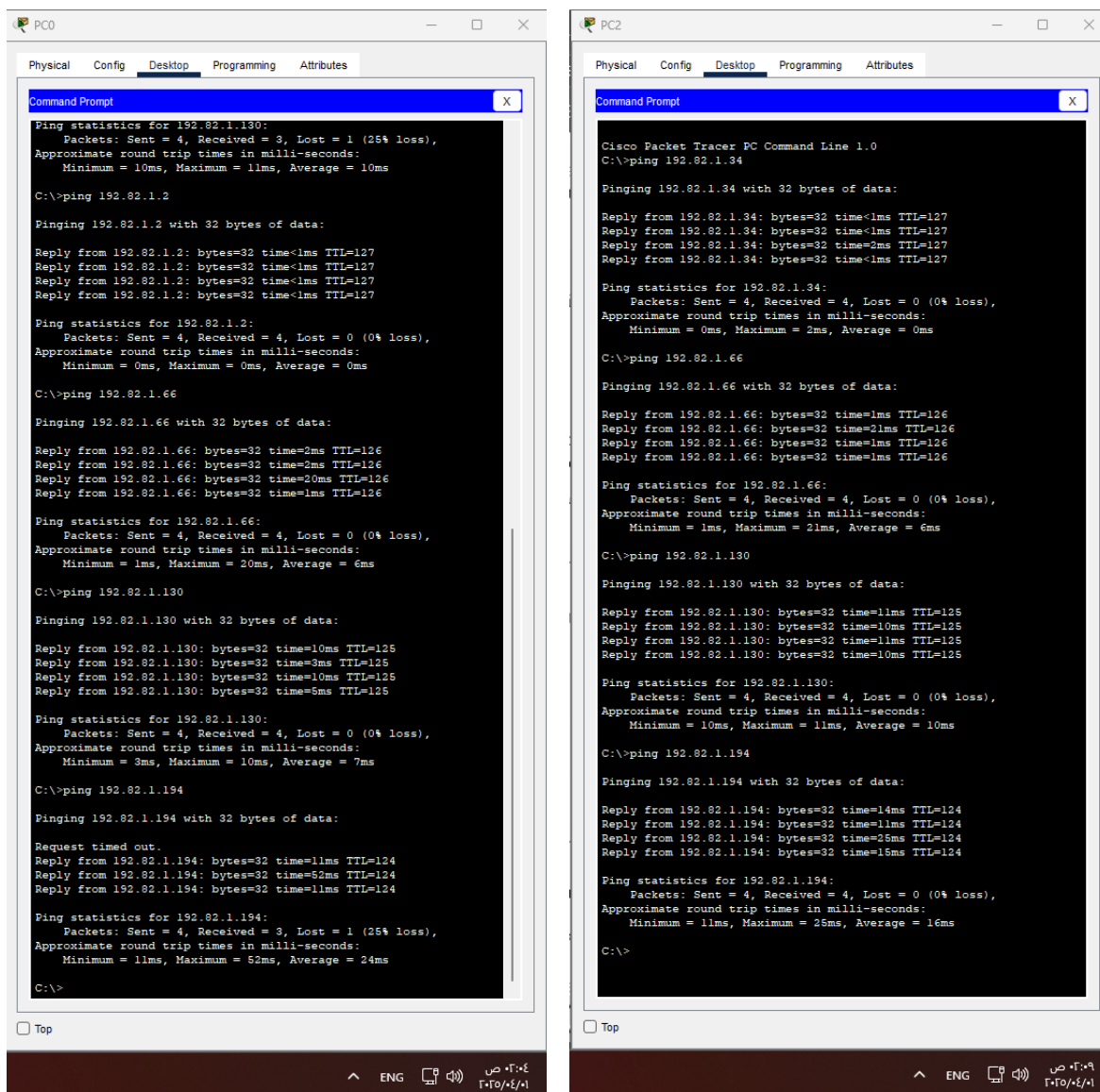


Figure 3-9: Connection from PC0, and PC2

Figure 3-10 illustrates the connection from PC1 to PC4 and from PC1/PC4 to PC5 respectively

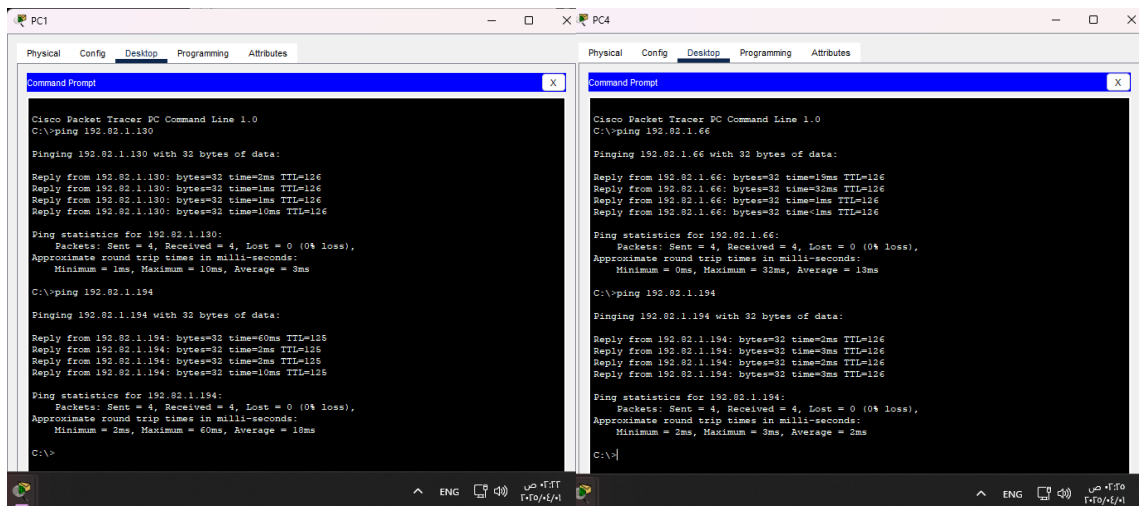


Figure 3-10: Connection from PC1 & PC4

- There is no need to perform the same on PC5 since that is implemented implicitly from the previous test.
- I didn't test connection to PC3 because it is on the same subnet as in PC4.
- From the previous tests we see that all PCs are connected to each other, which implies that our work is perfect.

4 Conclusion

By doing this experiment, I made a close relation with dynamic routing protocols and the processes involved in building a stable network. By installing and connecting various devices via both OSPF and BGP, I experimented with how the protocols distribute routing information, route redistribution among various routing domains, and the impact of administrative distances on routing decision-making. The practical experience of assigning IP addresses, forming neighbor relationships, and observing network behaviors using commands such as *ping*, *show ip route*, and *show ip bgp summary* greatly improved my troubleshooting capabilities. Furthermore, the use of packet sniffing software was important in transferring detailed insights in the details of BGP message types—OPEN, UPDATE, KEEPALIVE, and NOTIFICATION messages—and how they work towards network stability and efficiency. This task reinforced my technical abilities in network installation and protocol management. In addition, it highlighted the importance of careful planning, precise configuration, and continuous monitoring in ensuring flawless and seamless network communication and functionality.

5 References

[1] ENCS4130 Manual, "Computer Network Lab, EXP. No. 05– BGP," Accessed on: 3-Apr-2025.

[2] "Border Gateway Protocol," Wikipedia, 8 June 2016. [Online]. Available: https://en.wikipedia.org/wiki/Border_Gateway_Protocol. [Accessed: 3-Apr-2025].

6 Appendices

6.1 Appendix A: BGP Peers/Neighbors

BGP peer relationships allow routers to exchange critical routing information and maintain accurate routing tables. There are two types of BGP neighbor relationships:

- iBGP Peers: These are BGP neighbors within the same AS. iBGP is used to exchange routing information between routers that belong to the same AS. iBGP routes have an Administrative Distance (AD) of 200, making them less trustworthy compared to EIGRP, OSPF, and RIP routes. [1]
- eBGP Peers: These are BGP neighbors between different ASes. eBGP facilitates routing information exchange across ASes, typically over the Internet. eBGP routes have an AD of 20, making them more trustworthy than internal routes. [1]

Figure 6-1 “illustrates AS connectivity with BGP links. Routers B, C, and D, located in AS 200, can form an iBGP peer relationship among themselves. Meanwhile, Router A in AS 100 and Router B in AS 200 establish an eBGP peering.” [1]

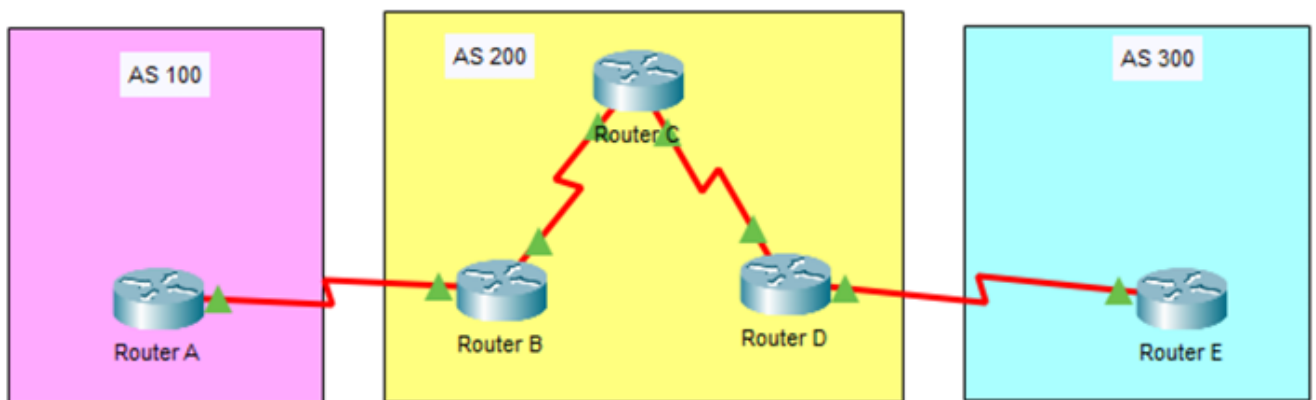


Figure 6-1. Example of AS connectivity with BGP links.[1]

6.2 Appendix B: BGP Peers Messages

BGP make peer relationships and through a set of messages, each serving a specific function. These messages include:

- **OPEN Message:** Exchanged between peers to initiate a BGP session. It contains key parameters required for session setup:
 1. **BGP Version:** Must match between peers for compatibility. [1]
 2. **Local AS Number:** Identifies the AS number of the originating router. [1]
 3. **BGP Router ID:** A unique identifier assigned to the router.[1]
- **KEEPALIVE Message:** Sent periodically (default: every 60 seconds) to confirm that the peer is still active. If a router does not receive a KEEPALIVE message within the Hold-Time (default: 180 seconds), the remote peer is considered unreachable. [1]
- **UPDATE Message:** Used to exchange routing information between peers. It contains details about network reachability and updates to the routing table. [1]
- **NOTIFICATION Message:** Sent when error occurs in the BGP session. Upon receiving a NOTIFICATION message, the BGP session is immediately terminated and must be re-established. [1]

6.3 Appendix C: BGP Finite-State Machine (FSM)

The BGP session goes through several states, with each state representing a distinct stage in the connection process.

- “Idle: This is the initial state of the BGP session. BGP waits for a start event (e.g., manual configuration or a policy trigger). When a start event occurs, BGP moves to Connect state.” [1]
- “Connect: In this state, BGP actively initiates a TCP connection to the peer. If the three-way TCP handshake completes successfully: (i) BGP resets the ConnectRetryTimer, (ii) Sends an OPEN message to the peer, and (iii) Moves to the OpenSent state. If the handshake fails (e.g., no response from the peer) or the ConnectRetryTimer expires, BGP transitions to Active state.” [1]
- “Active: BGP waits for an incoming TCP connection request from the peer. If a connection request arrives: (i) BGP establishes the session, (ii) Sends an OPEN message, and (iii) Moves to OpenSent state. If this attempt for TCP connection fails: (i) BGP resets the ConnectRetryTimer, and (ii) Moves back to Connect state.” [1]

- “OpenSent: BGP has sent an OPEN message and is waiting for the peer’s OPEN message. Once received, the router validates the message parameters (e.g., BGP version, AS number, etc.). If valid, a KEEPALIVE is sent, and the state moves to OpenConfirm. If errors exist, a NOTIFICATION is sent, and BGP resets to Idle.” [1]
- “OpenConfirm: BGP waits for a KEEPALIVE or NOTIFICATION message. If a KEEPALIVE is received, the session transitions to the Established State. If a NOTIFICATION is received or the Hold Timer expires, BGP resets to Idle state.” [1]
- “Established: The session is fully functional, and peers exchange UPDATE messages to share routing information.” [1]